

Project 2: Chest X-Ray - Pneumonia Detection with VGG CNN

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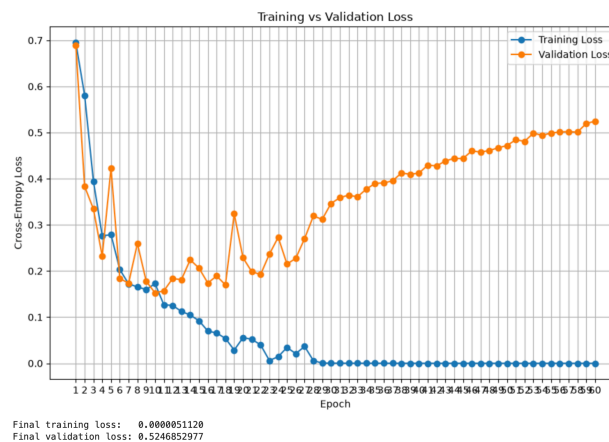
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A VGG-style Convolutional Neural Network was implemented from scratch using PyTorch to classify given chest X-rays as NORMAL or PNEUMONIA. The feature extractor follows the specified requirements i.e. three blocks (1->64, 64->128, 128->256 channels), 3x3 convolutions with padding 1, ReLU after every convolution, and a MaxPool2d (2, 2) closing each block. Stratified split was used. The given *trainval* folder of 1,234 images was divided 80/20 into 987 training and 247 validation images.

A. Learning rate, batch size and epochs

The model was trained with the Adam optimiser and a learning rate of 0.0001 was selected. The batch size was set to 8, for 60 epochs. These hyperparameters were selected by manually experimenting with different values and combinations and comparing the validation performance. A larger learning rate of 0.001 trained faster but produced a noisier, less stable validation curve and sometimes even caused the model to underfit, whereas 0.0001 gave a smoother descent. The small batch size of 8 gave better validation performance, and 60 epochs were enough for the training loss to fully converge and show the complete learning behaviour.

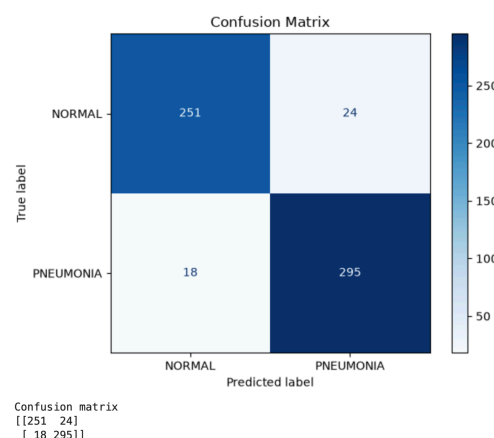
B. Training and validation loss curves



C. Training, convergence and overfitting

From *Figure 1*, the training loss can be observed to fall steadily and converges to approximately 0.000005 by epoch 28 and stays flat afterwards. This means the model fully converged on the training set and is classifying every training image with confidence. On the flip side, validation loss reaches its minimum of about 0.15 near epoch 10 and then starts to rise steadily to about 0.52 by the final epoch. It is evident from the figure that the model has started to overfit. Training loss falling to near-zero while validation loss climbs is the direct evidence of this. It is worth noticing that the final model still generalises well on the unseen test set at 92.9% accuracy.

D. Confusion matrix on the test set



On the test set, the model achieves precision of 0.9331 and recall of 0.9127 for the NORMAL class, and precision of 0.9248 and recall of 0.9425 for the PNEUMONIA class, and an overall accuracy of 0.9286 . All values exceed the required 0.91 target. The matrix shows 251 true negatives, 295 true positives, 24 false positives and 18 false negatives.

E. Importance of precision and recall over accuracy

Accuracy is just a number that combines every prediction together and treats both types of error equally. This hides the two important things. First, the classes are imbalanced i.e. 313 pneumonia against 275 normal in the test set. Even if a model simply predicted PNEUMONIA for everything, it would still score approximately 53% accuracy while being clinically useless. Second, a medical diagnosis needs the two error types to be judged separately. Precision and recall report performance for each class individually. While recall shows how many true pneumonia cases were caught, precision shows how often a pneumonia prediction is correct. Accuracy can let a strong score on one class overshadow a poor score on the other, which is exactly why these specific metrics are more informative than overall 'Accuracy'.

F. Meaning of a false positive and a false negative for a patient

A 'False Positive' implies that a healthy patient who is NORMAL is predicted as PNEUMONIA. While there may be no direct harm to their health, the patient is wrongly told they may be ill, leading to unnecessary anxiety, further tests and possibly unneeded treatment. Whereas a 'False Negative' is a patient who is genuinely sick and has PNEUMONIA, but the model predicts it as NORMAL. This patient is told they are healthy and sent home untreated which means their pneumonia remains undetected and untreated which can become life-threatening in future. A false negative result is the far more dangerous of the two errors in this case.

G. Safety and readiness as a real-world diagnostic aid

In my opinion, this model can be deployed in a real-world scenario but strictly as a diagnostic aid. The model should not be deployed as a standalone autonomous diagnostic tool. The most important weakness is the 18 false negatives out of 313 pneumonia cases, which means that roughly one in eighteen sick patients would be missed and remain untreated. For a potentially fatal condition, this error rate is unacceptable for autonomous use. The 24 false positives would still be considered less dangerous but would still generate unnecessary follow-up. A responsible use case would be as an assistive screening aid that flags pneumonia for a radiologist to confirm and a human always making the final decision. Furthermore, the images are only 28x28 pixels, and the dataset is small compared to real world settings, so performance on diverse real-world populations and equipment is uncertain. Gathering more, high resolution data and adding regularisation to reduce the overfitting, which was observed in *Figure 1*, would be necessary first steps.

H. Architectural advantage of consecutive 3x3 convolutions

Using several consecutive 3×3 convolutions instead of a single larger convolution (e.g. 5×5) gives the same effective receptive field but with two key advantages. First, it uses fewer parameters. The number of weights in a convolution grows with the square of its kernel size, so one large kernel would be more expensive, whereas a stack of small 3×3 kernels covering the same area needs fewer weights in total which helps reduce computation and the risk of overfitting. Second, it adds more non-linearity. Each 3×3 convolution is followed by its own *ReLU*, so stacking them inserts several non-linear steps where a single large convolution would only have one. This helps the network learn a richer, more complex function from the same region of the image. This is the core design principle of the VGG architecture.